Khue (Cherry) Pham

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Education

Northeastern University, The Roux Institute – M.S. Computer Science Dec 2027

Olin College of Engineering – B.S. Electrical and Computer Engineering May 2025

Technical Skills

Languages: Python, SQL

Software Engineering: AWS, Git

ML/AI: PyTorch, Fine-tuning

Experience

Florida Center for Reading Research Remote

Machine Learning Engineer Sep 2024 – Aug 2025

- Built a PyTorch/LangChain pipeline and fine-tuned a LLaMA-based LLM on early literacy corpora, evaluating prompting and fine-tuning variants against rubric-aligned benchmarks to select version for cross-state pilot deployment.
- Spearheaded a domain-expert integration strategy, identifying and onboarding subject matter experts to fill critical research gaps and accelerate model development cycles.
- Built RAG pipelines with pgvector-backed semantic search (PostgreSQL), improving retrieval accuracy by 15–25% on children's reading development tasks; evaluated using BLEU/ROUGE benchmarks.

Air Partners Needham, MA

Data Engineer Jun 2023 – Aug 2023

- Designed a normalized data model and modular ETL pipeline to integrate time-series data from 18 disparate IoT sensors into an analysis-ready dataset, processing ≈3 million records across 2 years of hourly readings; reduced pipeline latency by 30% through parallelized processing and optimized API batching.
- Consolidated scattered Python and R scripts into a reproducible R workflow with documentation, enabling non-technical team members to modify and rerun the pipeline independently.
- Prepared 4 data visualizations translating atmospheric sensor data into actionable insights supporting environmental justice litigation by the Roxbury community group.

Olin College – College as a Living Lab Needham, MA

AI Pedagogy Researcher Jun 2025 – Aug 2025

- Evaluated prompting strategies across 4–6 model iterations against professor-graded rubrics, reaching 80–87% alignment with human graders through iterative prompt design and layer-specific fine-tuning.
- Led institutional research into emerging AI tools to synthesize evidence-based policy recommendations for responsible LLM integration across college curricula and operations.

Projects

ENCoM – Real-Time Generative Speech-to-Speech System | GitHub

- Designed and ran experiments comparing a Mimi+Mamba hybrid architecture against a controlled cascaded baseline, achieving 2.79x inference speedup while maintaining audio quality (MOS) on Switchboard/Expresso evaluation sets.
- Conducted ablation studies on architecture variants, measuring perplexity for same-condition comparisons and automated speech-quality scoring via Whisper transcription for cross-condition evaluation.
- Benchmarked end-to-end performance against SOTA cascaded pipelines (Whisper/GPT/Kokoro), delivering a 3x improvement in RTFx and significantly higher barge-in recovery precision.

Publications & Research

Co-authored and co-presented 2 peer-reviewed papers at **Frontiers in Education (FIE)**, a top-ranked engineering education conference, on game-based learning's effects on student motivation and academic performance.